

Synthetic Market Phase 1

First-pass manifold findings for the controlled market-only synthetic dataset. Research anchors: MARKET_COUNTING_MANIFOLDS_PLAN.md, MARKET_MANIFOLD_RESEARCH_PLAN.md, and MARKET_MANIFOLD_IMPLEMENTATION_PLAN.md.

DATE	PROMPTS	LAYERS	VOLUME
20 March 2026	510 total, 170 market-only	48	Dedicated synthetic

MAIN READ

Phase 1 succeeded as controlled variable isolation — but not yet as manifold discovery.

The synthetic row states preserve clean latent variables extremely well (R^2 up to 0.990). But the scalar sweeps do *not* yet look like a crisp counting-manifolds-style one-dimensional geometry. The manifold story is only partial at this stage.

TOTAL PROMPTS	CAPTURED SLICE	BEST LATENT R^2	BEST SCALAR GEOM.
510 170 market-only	170 15 structure keys	0.990 edge_after_fee L10	0.659 net_flow Spearman L28

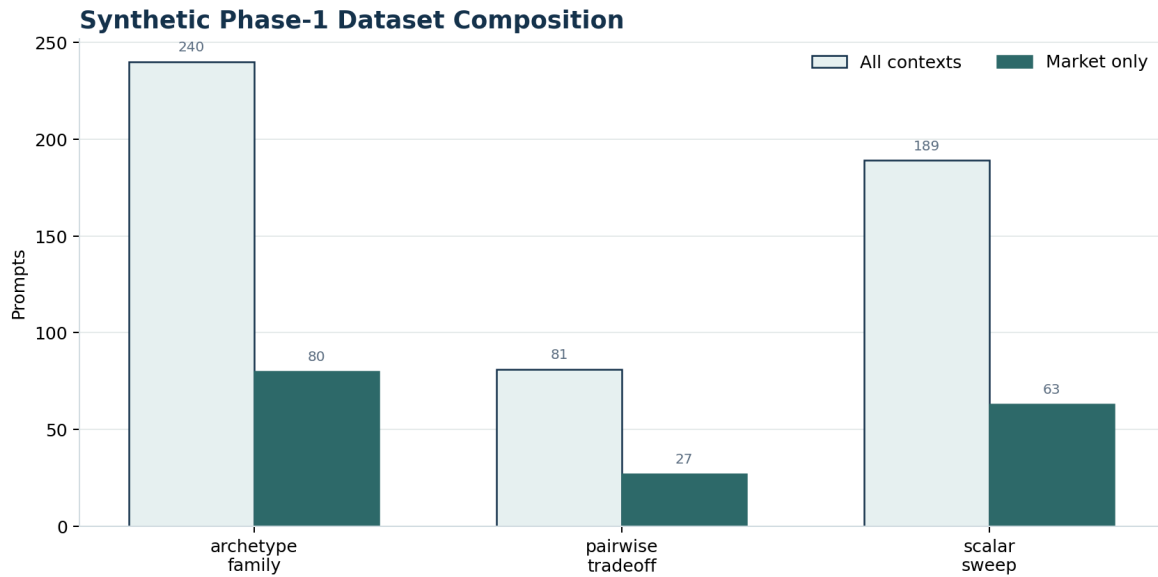
Scope

This phase intentionally strips the problem down. The goal was *not* to prove a rich market manifold in one step. The goal was to isolate clean latent variables, capture the synthetic prompts end to end, and ask whether those variables are linearly present in the market-row states before scaling into denser sweeps and context ladders.

- **Neutral assets.** The prompts use synthetic assets A/B/C/D to remove token-identity priors from real DX data.
- **Clean latent variables.** Labels are constructed directly for attractiveness, risk_adjusted, edge_after_fee, and pairwise preference.
- **Market-only first.** Phase 1 focuses only on the market slice before settings or portfolio context are layered back in.
- **Full activation path.** Prompts were captured on a dedicated Modal volume, pooled into row and section states, and analyzed on held-out splits.

Dataset

The full synthetic phase-1 dataset contains 510 prompts, but this report covers the 170 market-only captures that isolate the market representation before policy overlays. Prompt families: `scalar_sweep`, `pairwise_tradeoff`, and `archetype_family`. Four asset rows per prompt, yielding 680 unique market-only row states. Each captured prompt was reduced into `row_mean_i`, `row_eos_i`, section states, and `last_token`.



Phase-1 prompt counts by family. The report slice uses market-only prompts; the full dataset already includes low-risk and high-risk context-ladder variants for the next phase.

Main Quantitative Findings

Question	Best State	Metric	Interpretation
attractiveness_score	row_mean L10	R ² 0.989	Clean latent score almost perfectly recoverable from row states.
risk_adjusted_score	row_mean L10	R ² 0.990	Risk-adjusted valuation preserved without needing late context.
edge_after_fee_score	row_mean L10	R ² 0.990	The model retains a highly usable notion of fee-adjusted edge.
is_best_asset	row_mean L1	AUROC 1.000	Best-asset choice is trivial on this controlled slice.
A beats B attractiveness	diff: row_mean L0	AUROC 1.000	Pairwise preference immediately separable from row-difference vectors.
A beats B risk-adjusted	diff: row_mean L3	AUROC 1.000	Risk-adjusted pairwise preference also perfectly separable.

This is the most important split in the report: probe results are *extremely* strong, but scalar geometry results are only *moderately* strong. The model preserves the variables, but whether it organizes them as a simple low-dimensional manifold is not yet clear.



Held-out regression probes across layers. `row_mean` dominates and peaks around layer 10 for all three latent scores.

Choice-oriented probes are nearly perfect on this controlled slice — useful for pipeline validation, but confirms Phase 1 is an easy synthetic setting.

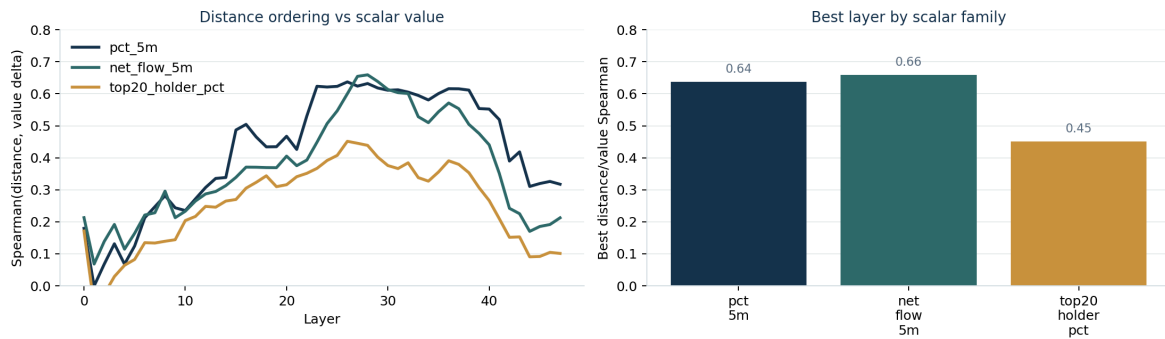
Geometry Read

The scalar sweep analysis is where the manifold question starts to become nontrivial.

Scalar Family	Best State	Layer	Dist/Val Spearman	Participation Ratio
pct_5m	row_mean	26	0.637	3.09
net_flow_5m	row_mean	28	0.659	3.70
top20_holder_pct	row_mean	26	0.452	3.86

- The scalar sweeps are *ordered enough* that activation distances track scalar differences.
- But the best correlations are well below 1.0.
- Participation ratios around 3–4 are higher than the clean intrinsic-dimension-1 story needed for a counting-manifolds-style result.
- `pct_5m` and `net_flow_5m` look more coherent than `top20_holder_pct`.

Scalar Sweeps Show Partial Geometry, Not a Crisp 1D Manifold Yet



Scalar sweep geometry is only partially organized. `pct_5m` and `net_flow_5m` show moderate ordering; `top20_holder_pct` is weaker. The overall structure is substantially higher-dimensional than a clean 1D curve.

Interpretation

SUPPORTED NOW

The synthetic prompts successfully isolate clean latent variables, and those variables are strongly present in the row states.

NOT SUPPORTED YET

We do not yet have a compelling one-dimensional market-counting manifold analogous to the linebreak paper.

Best current interpretation:

- Phase 1 proves that the capture stack, pooled synthetic structure path, and analysis stack are all working end to end.
- The model linearly preserves attractiveness, risk_adjusted, and edge_after_fee with very high fidelity.
- Pairwise preference is easy enough in this slice that the relevant differences are already perfectly separable.
- The geometry result is the limiting one: there is *some* scalar ordering, but not yet the crisp low-dimensional scalar manifold we would want for a stronger counting-manifolds analogue.

What Phase 1 Accomplished

Relative to the counting-manifolds-inspired plan, Phase 1 completed the necessary groundwork:

- Synthetic prompt generator with neutral assets and clean labels.
- Neon tables and views for synthetic examples, asset rows, and pairwise rows.
- Dedicated Modal volume for synthetic captures, separate from the crowded xenon-data volume.
- Smoke-tested and full market-only capture run.
- Synthetic structure pooling over the full captured slice.
- First-pass manifold analysis with held-out probes and scalar geometry metrics.

What Is Still Missing

Phase 1 does *not* yet close the main research question.

- The scalar sweeps are still too coarse: 21 values per family with effectively one market-only prompt per value.
- Backgrounds are too fixed to cleanly distinguish invariant scalar geometry from prompt-template artifacts.
- The current phase is market-only; it does not yet show how settings or portfolio context twist the synthetic manifold.
- The current phase does not include causal interventions or row-level patching.

Next Phase

The most defensible next move is to expand the synthetic market-only geometry phase before writing strong manifold claims:

1. Increase scalar density substantially and repeat each scalar value across multiple background rosters.
2. Add a stripped-down ultra-minimal scalar sweep where only one asset row changes and everything else is flat.
3. Rerun the same geometry analysis and look for stronger monotonicity, lower participation ratio, and more stable layer-local organization.
4. Only then add the context ladder back in to see whether settings twist or reweight the synthetic market geometry.

If those denser sweeps still fail to produce cleaner scalar geometry, that would be the first real challenge to the current hypothesis.

Synthetic Market Phase 1 Report — 20 March 2026. 170 market-only captures from 510 total prompts. Qwen3-30B-A3B surrogate, dedicated synthetic Modal volume, 48 layers.